Table 2.2: Difference between SEM and TEM

S.no	Properties	SEM	TEM
1.	Types of electrons	It is based on scattered electrons that are emitted from the surface of a specimen	It is based on transmitted electrons
2.	Sample preparation	Sample can be of any thickness and is coated with a thin layer of a heavy metal such as gold or palladium and mounted on an aluminum slab	Laborious sample preparation is required. The sample has to be cut into thin section s so as to allow electrons to pass through it and are supported on TEM grids
3.	Resolution	The resolution is up to 20nm	TEM has much higher resolution than SEM. It can resolve objects as close as 1nm
4.	Magnification	The magnifying power of SEM is up to 100,000X	The magnifying power of TEM is up to 5,000,000X
5.	Image formation	SEM provides a 3 dimensional image. Secondary or back scattered electrons are captured, detected and displayed on computer screen	TEM provides a 2 dimensional image. Transmitted electrons hit a fluorescent screen giving rise to a shadow image. The image can be studied directly by the operator or photographed with a camera
6.	Application	SEM is used to study the topography and atomic composition of specimens	TEM is used to study the interior of cells, the structure of protein molecule, the organization of molecules in viruses and cytoskeletal filaments and the arrangement of protein molecules in cell membranes

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